

# KernelAscent: The Complete Report

Separating search, selection and learning in verified GPU-kernel self-optimization — does *persistent* self-improvement compound? (results, mechanism, and intuition)

KernelAscent

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## Abstract

KernelAscent asks a question that raw capability benchmarks cannot: not *how good* a model is at writing GPU kernels, but whether it can *make itself better* at writing them — and whether that improvement *compounds*. We built a five-rung ladder of recursion (one-shot capability; weight-level self-training; frozen-weight procedure self-editing; a closed model rewriting an open trainee’s harness; and open-ended self-play where a model authors the very frontier it must then beat) and ran it for three weeks on a 72-GPU cluster across model scales from 0.5B to 14B and a panel of frontier API models. This report is the full synthesis: the design rationale for each rung, every quantitative result with its uncertainty, the mechanistic account of *why* verified self-training does not compound in this domain, and the accumulated intuition and operational lessons of the run. Our headline is a bounded null, analysed at the level of the independent replicate (the trajectory, not the round): at matched compute, carrying a self-training lineage forward does not beat a matched reset (TOST-equivalent at  $\delta=0.05$  over 57 trajectories,  $\text{BF}_{01}=5.6$  — *moderate* evidence for the null, robust to restricting to full-length runs), while verified *search* at the same budget does strictly better ( $-0.111 [-0.138, -0.083]$ , CI excludes zero) — localizing the failure to a discovery→assimilation bottleneck rather than to an absence of capability. **We are precise about which claim we test.** *Persistent* self-improvement asks whether inherited updates improve future performance; *recursive* self-improvement asks whether those updates improve the system’s ability to generate its own future training signal or curriculum. This report tests persistence directly (lineage vs. reset) and finds it equivalent to a reset. The only rung that probes recursion — the recursion-interruption fork, in which the producer is frozen at a checkpoint while its descendant continues — returns a one-time producer upgrade, not compounding. The self-play rung, which was designed to test recursion through an endogenous curriculum, did not reach a testable state: its live authors failed to yield enough accepted, valid tasks for  $L - F$  to be defined. We therefore make no recursive-self-improvement claim in either direction, and we do not treat our persistence null as evidence about recursion.

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# 1 Thesis: recursion, not capability

The central intellectual commitment of KernelAscent is that the interesting axis of “self-improvement” is **recursion**, not the raw height of a single-shot score. *What this report actually measures is the weaker of the two claims — persistence —* because that is the one the data can support; the recursion rungs either return a one-time upgrade (the fork) or never became measurable (self-play). We keep the framing because it is what the instrument was built to falsify, and we mark throughout which claim each result licenses. A model that writes an excellent kernel once has demonstrated capability; a model whose *next* kernel is better *because of* the last one has demonstrated something qualitatively different. Most benchmarks conflate the two because they measure a fixed model against a fixed test. We instead measure the *derivative*: does an improvement, once made, become part of the improver, so that improvements accumulate?

GPU-kernel optimization is an unusually clean substrate for this question. Correctness is checkable against an fp32 reference with no ambiguity; speed is measurable against a roofline; the task space is procedurally generable at controlled difficulty; and there is a genuine, unbounded frontier (there is always a faster kernel). A verifier this crisp removes the usual confound of a noisy reward, so any failure to compound cannot be blamed on the grader.

**Intuition.** *If self-improvement is real anywhere, it should be visible where the reward is a clean, programmatic verifier and the frontier is real. Kernels are that place. So a null here is informative in a way a null on a fuzzy task would not be: it cannot be explained away by reward noise.*

## 2 The RSI ladder: five rungs of recursion

Each rung isolates a different *channel* through which improvement could feed back into the improver, in increasing order of recursion depth. The design principle throughout is the *matched control*: every “recursive” arm is paired with a non-recursive arm at identical compute, so a positive result cannot be attributed to simply spending more.

**T1 — Capability (one shot).** Can the model write a correct, fast kernel in a single attempt? No recursion; this is the baseline skill that gates everything above it. Metric: correct rate, fast rate, mean bounded attainment  $C$ .

**T2 — Weight-RSI (the title claim).** Across rounds, the model generates kernels, keeps the verified-correct ones, and trains its own weights on them, then is re-scored on a held-out split. Because the weights change, the *improver itself* changes. The decisive control is **lineage vs. matched reset**: does carrying the accumulated weights forward beat restarting from base each round at equal compute? Metric:  $L - R$  per round. Open-weight only.

**T3 — Procedure-RSI (recursion without weights).** A frozen-weight model edits its own *executable research procedure* — a strategy library, a solver prompt, and an archive of verified kernels — from the evidence of the last round. This is how a closed API model can sit on the RSI axis at all. Metric: held-out procedure quality  $Q$ , gain vs. the frozen procedure.

**T4 — Closed→Open (improvement through tooling).** A closed frontier “researcher” rewrites the *training harness* (data selection, LoRA hyperparameters, curriculum) of an open-weight trainee. The closed model never changes its own weights; the improvement is transmitted through the tooling it authors. Metric: open-trainee  $C$  under the improved vs. frozen harness.

**T5 — Self-play (the true-RSI metric).** The model *authors its own strictly-harder tasks* and solves-and-improves on them, so difficulty and capability co-evolve. To separate “a harder curriculum” from “a co-evolving author” we run three arms at equal budget on one fixed held-out ladder: **S** (static frontier), **F** (escalating frontier, frozen author), **L** (escalating frontier, live/evolving author). The primary signal is  $L - F$  — the extra gain from letting the author evolve — with anti-reward-hacking gates rejecting degenerate self-authored tasks.

**Intuition.** *The ladder is deliberately ordered by how “expensive” the feedback channel is to fake. A one-shot score (T1) says nothing about recursion. A weight update (T2) is the strongest possible feedback — it modifies the model itself — so it is the title claim. Procedure editing (T3) and harness authoring (T4) are weaker, externalized channels that a frozen model can still use. Self-play (T5) is the only rung where the problem distribution itself is endogenous, which is the real definition of open-ended improvement. We expected the strongest evidence at T2 and the most fragile at T5; the data inverted our expectation about where gains would even appear.*

### 3 Infrastructure: three weeks on 72 GPUs

The experiments ran on a 9-node, 72-GPU cluster (A100-class), continuously for three weeks. A few infrastructure decisions shaped what was learnable, and several operational lessons are worth recording because they materially affected the results.

**Compute discipline.** Every rung’s recursive arm is compute-matched to its control by a full ledger that counts generation *and* verification *and* training tokens/steps. Without this, “self-training helped” is unfalsifiable — it may just be more sampling. The matched ledger is what turns T2 from a demo into a test.

**Resumability.** All runs checkpoint to S3 on a 300s daemon so an instance death never loses more than one interval, and pods push independently of the operator’s credentials. This is what made a three-week run survive tunnel drops, credential expiry, and node churn.

**Operational lessons (recorded because they bit us).**

- **Grading dominates wall-clock.** Crash-isolated per-kernel grading (each candidate graded in its own process so a segfaulting kernel cannot take down the harness) costs  $\sim 12$ s/kernel. For coverage-style experiments requiring hundreds of grades per evaluation, a single round can exceed 2.5 hours — which made the randomized coverage-transplant experiment *computationally infeasible* and forced us to establish the coverage mechanism more cheaply from  $p$ -maps and forensics instead.
- **Large-model rounds are slow.** A 7B compounding round costs  $\sim 64$  minutes. At six rounds per seed that is  $\sim 6.4$  hours per seed, so the large-scale arms accumulate statistical power slowly and are sensitive to any run reset from node churn — the reason our 7B/14B equivalence is pooled-significant but not yet individually powered.
- **Measurement hygiene is a result-integrity issue.** We twice caught inflated counts caused by stale local caches masking a silently-failed sync, and once caught a “fleet collapse” that was merely a dropped port-forward tunnel hiding two dozen busy GPUs. Every headline number in this report is taken from a *verified* re-read of canonical storage, not a convenience cache.

- **A single silent bug can fabricate a finding.** An early “compounding collapse” (both train and held-out  $C$  dropping to exactly 0.000 after one SFT step) was a generation-after-SFT bug, not a scientific null; a zero-update adapter evaluated correctly, which is how we distinguished the two. Pre-fix runs are therefore *uninterpretable* and are excluded from every statistic by a canonical-seed allowlist.

**Intuition.** *The most dangerous failure mode in an autonomous, long-horizon run is not a crash — it is a plausible wrong number. Crashes announce themselves; a stale cache or a contaminated seed quietly shifts an estimate. The single highest-leverage habit of the whole run was refusing to trust any number that had not been re-derived from canonical storage with the broken seeds excluded.*

## 4 Results by rung

### 4.1 T1 — Capability

On the E0 gate (15 fixed Medium procedural tasks,  $k=5$ , 75 trials/model, 95% Wilson CIs), capability separates into *two walls*: correctness ranks the low/mid band, while *speed* (**fast\_rate**, the fraction beating the `min(eager, torch.compile)` roofline by  $\geq 1.10\times$ ) is the frontier discriminator.

| Model                   | correct | fast  | mean $C$ |
|-------------------------|---------|-------|----------|
| GPT-5.6 Terra (api)     | 0.920   | 0.667 | 0.793    |
| GPT-5.6 Sol (api)       | 0.907   | 0.600 | 0.753    |
| Kimi K2.5               | 0.387   | 0.240 | 0.313    |
| Fable 5.1               | 0.360   | 0.227 | 0.293    |
| Palmyra-X5              | 0.533   | 0.040 | 0.287    |
| gpt-oss-120b            | 0.467   | 0.080 | 0.273    |
| Qwen2.5-Coder-7B (open) | 0.280   | 0.000 | 0.140    |

Table 1: T1 capability (top rows). *Every* open model  $\leq 14\text{B}$  has **fast\_rate** = 0: they can sometimes write a correct kernel but essentially never a frontier-fast one; Coder-14B and Gemma-3-27B additionally hallucinate `torch` internals. Source: `leaderboard.json`.

**Intuition.** *The two walls matter for everything above: T1 shows that below the frontier, models are correctness-limited long before they are speed-limited. So the RSI rungs are mostly testing whether a model can compound its way up the correctness wall, not whether it can shave microseconds — which is why “where  $C$  moves, it moves via correctness acquisition, not speed” recurs throughout.*

### 4.2 T2 — Weight-RSI: a rigorously-powered compounding null

The title claim is where the strongest evidence lives, and it is a *null*. Over multi-round rejection-sampling SFT, carrying the lineage forward does not beat a matched reset.

**Finding.** The independent replicate is the *trajectory* (one seed, one base checkpoint, one held split, one accumulated adapter), not the round: rounds within a run are serially dependent, and an earlier draft of this report pooled them as if independent. Pooled across scales (0.5B–14B) over **57 trajectories** comprising 228 round-comparisons,  $L-R = -0.010 [-0.035, +0.016]$ , TOST-equivalent at  $\delta=0.05$  with  $\text{BF}_{01}=5.6$  — *moderate* evidence for the null. A cluster-robust estimate clustered on trajectory agrees ( $-0.002 [-0.024, +0.020]$ ,  $\text{BF}_{01}=7.4$ ). The naive round-level pooling

would have reported  $\text{BF}_{01}=14.7$ ; that figure is an artifact of the clustering and we do not claim it. The verdict is robust: restricting to trajectories with  $\geq 3$  rounds ( $n=36$ ) gives  $\text{BF}_{01} \approx 5.9$  and to full-length  $\geq 6$ -round trajectories ( $n=29$ ) gives  $\text{BF}_{01} \approx 4.9$ , both EQUIV.

| Scale              | traj. $n$ | rounds     | $L-R$ (95% CI)                              | TOST         | $\text{BF}_{01}$ |
|--------------------|-----------|------------|---------------------------------------------|--------------|------------------|
| Qwen2.5-Coder-0.5B | 12        | 51         | $-0.014 [-0.058, +0.030]$                   | EQUIV        | 2.6              |
| Qwen2.5-Coder-1.5B | 20        | 93         | $-0.001 [-0.037, +0.035]$                   | EQUIV        | 4.5              |
| Qwen2.5-Coder-3B   | 19        | 78         | $-0.027 [-0.086, +0.032]$                   | not equiv.   | 2.7              |
| Qwen2.5-Coder-7B   | 5         | 5          | $+0.024 [-0.126, +0.174]$                   | underpowered | 2.0              |
| <b>Pooled</b>      | <b>57</b> | <b>228</b> | <b><math>-0.010 [-0.035, +0.016]</math></b> | <b>EQUIV</b> | <b>5.6</b>       |

Table 2: Lineage vs. matched reset at the **trajectory** level (primary estimand; `equivalence_tost.py`).  $n$  counts independent trajectories; “rounds” counts the round-comparisons they contain. Intra-trajectory ICC=0.093 (design effect 1.28), so the naive  $n=228$  rounds carry the evidence of  $\approx 178$  independent observations. Individual scales are now only moderately powered — 3B is *not* individually equivalent once clustering is respected, and 7B ( $n=5$ ) is not interpreted. Per-round,  $L-R$  shows a *transient* early advantage (rounds 1–2) decaying to  $\approx 0$  by round 3.

The uncontrolled T2 boards (`rsi_leaderboard.json`, `tier_speed_rsi.json`) do show apparent per-model “compounding” — the strongest being **stable-code-3b**, self-fresh-frozen  $+0.226$ , sign-consistent across 2 seeds; and `starcoder2-15b`  $+0.074$  across 3 seeds. But the decisive recursion-interruption fork (A1, `fork.json`) dissolves it: for `deepseek-1.3b`, continuing to self-train vs. freezing the producer at the fork gives self-ckpt =  $-0.001$  — a **one-time upgrade, not compounding**. Where  $C$  does move it is almost entirely *correctness acquisition*, not speed (`decomp.json`: `deepseek-1.3b`  $\Delta_{\text{correct}} + 0.89$  vs  $\Delta_{\text{speed}} + 0.38$ ).

**Intuition.** *The lineage-vs-reset null is the cleanest single fact in the project, and it is stronger than “updates are small.” If updates were merely small, lineage and reset would both drift slowly but lineage would still inch ahead. Instead they are indistinguishable — which means the accumulated weight state carries no useful cumulative information beyond what one round of fresh self-training recovers. That is a statement about the state, not the step size.*

## An H100 replication under a correctness-reliability metric

The A100 compounding null above is measured on the headroom score. On H100 that score has no range (Section 6), so we re-ran the same protocol under a pre-registered second metric — *pass-rate*, the fraction of the  $k$  sampled candidates that verify (pre-registration, Amendment 1). Eight trajectories, 1.5B and 3B, two seeds per arm:

| Arm                                       | lineage – reset | 95% CI           |
|-------------------------------------------|-----------------|------------------|
| control (published protocol)              | +0.184          | [+0.080, +0.287] |
| inject (teacher coverage on failed tasks) | +0.184          | [+0.089, +0.278] |
| inject – control                          | +0.000          | —                |

Both CIs exclude zero: on this metric, accumulation beats a matched reset. This is the first non-degenerate compounding measurement we obtained on H100, and we report it with three limits that between them prevent it from supporting a compounding claim.

**First, the search baseline is degenerate under this metric and we do not report it.** Best-of- $N$ 's mechanism is the *maximum* over  $k(r+1)$  draws; pass-rate is a *mean*, which is invariant to the number of draws. Measured: as the budget grew five-fold,  $C_{\text{best-of-}N}$  moved  $-0.024$ . The apparent lineage – best-of- $N = +0.676$  (positive in 37/37 rounds) would therefore *reverse* our search-beats-training result using an artifact. **A baseline is only a baseline with respect to a scoring rule;** changing the rule silently changes what every comparator means, and each must be re-derived rather than carried over.

**Second, the metric saturates at the top.**  $C_{\text{lineage}} = 1.000$  in 18 of 37 rounds (49%) on a five-task held set. Once lineage pins at the ceiling the difference measures only how far *reset* fell below it, so the magnitude is a lower bound and the across-round trend is unreadable. Having diagnosed a floor effect and built a metric with range at the floor, we landed in a ceiling effect: **saturation is a property of the match between task difficulty and model ability, not of the metric,** and changing the metric only moves where the wall sits.

**Third, the contrast is not training-compute matched.** Lineage retains its adapter and trains every round ( $R \times$  steps); reset re-initialises and trains on one round's data ( $1 \times$  steps). Under a metric scoring reliability of correctness, “more supervised fine-tuning on verified-correct outputs raises the rate of correct outputs” is close to tautological. The result is evidence that accumulation improves correctness *reliability*; it is not evidence of recursive self-improvement.

Finally, the positive control did not fail — it ran out of work. Injected-task counts per round run  $[15, 2, 1, 0, 0]$  and  $[19, 3, 0, 0, 0]$ : injection targets tasks the student failed, and by round 2 it solves everything, which is why the inject–control delta is exactly zero. A five-task held set that a 1.5B model exhausts in two rounds cannot support a compounding claim under any scoring rule. The remedy is the 456-task generated bank (Table 6), not a third metric.

### 4.3 T3 — Procedure-RSI

When a frozen-weight model rewrites its own executable procedure, frontier models improve it substantially (`trackc.json`): GPT-6 Astra  $Q$   $0.298 \rightarrow 0.854$  (**+0.556**, growing over 6 rounds), Claude Sonnet 5 **+0.401**, Mistral Large 3 **+0.310**; a frozen-procedure control (Table-5.1) sits at  $+0.009$ . But the self-modify mechanism (`selfplay_mech.json`) shows the gain is **68% realized at round 0** (one-shot), **75% non-recursive**, with the archive saturating by round 1–2 and strategy sets already diverse (Jaccard 0.12).

**Finding.** The only models with *sustained* per-round procedure gain (Astra, Mistral) are the ones that started from *low*  $Q_0$  — i.e. they are consuming headroom, not demonstrating recursive self-insight. DeepSeek-V3.2's  $-0.234$  “self-degrade” is a 1-round interrupted-run artifact and should not be read as a finding.

**The one-shot reading is confounded by the harness, and we retract it.** The procedure is a list of strategy strings, and the harness asks the model for “ $\leq 12$ ” of them while displaying only 12 back. It is a *soft* request — the stored list is never truncated — so what it actually measures is instruction compliance. Frontier models comply exactly:

| Model           | Strategies per round   | Total $\Delta Q$ |
|-----------------|------------------------|------------------|
| GPT-6 Astra     | 12, 12, 12, 12, 12, 12 | +0.556           |
| Claude Sonnet 5 | 12, 12, 12, 12, 12, 12 | +0.401           |
| Mistral Large 3 | 12, 12, 12, 12, 12, 12 | +0.310           |
| GPT-5.6 Sol     | 12, 12, 11, 12, 11, 11 | —                |
| GPT-5.6 Terra   | 10, 12, 12, 12, 12, 11 | —                |

Five of five interpretable runs sit at 11–12 from round 0; four of five never grow at all. A procedure pinned at its limit in round 0 *cannot be observed to keep improving*, so “the model stopped improving its procedure” and “the harness stopped letting it” are not separable in this design, and the one-shot interpretation is not earned by these data.

Re-running the same harness with an open model at caps {12, 48, unbounded} shows procedures *do* grow when permitted — 6 of 6 clean runs grow, two exceeding 12 strategies — confirming that the cap binds something real. It does **not**, however, produce a dose-response in quality: on the clean runs mean  $\Delta Q$  is  $-0.046$  at cap 12,  $+0.021$  at cap 48 and  $-0.001$  unbounded ( $n=2$  per cell). Lifting the cap lets the procedure grow without making it better.

We flag a correction here rather than silently reporting the clean figures. An earlier version of this analysis reported a monotone rise to **+0.482** unbounded. That came from runs whose data directory sat on an inode-exhausted filesystem, where the reference build raised `EDQUOT` and the grader returned `False` for every candidate: those runs have  $Q_0 = 0.000$  and four to five exactly-zero rounds, so their apparent “gain” was recovery from a spurious zero baseline. We had already documented that failure mode (it is defect 5 in Table 5) and used the affected runs regardless. The exclusion is now enforced in the analysis script rather than left to memory: any run with  $Q_0 = 0$  or two or more exactly-zero rounds is dropped, loudly.

We state the limits plainly. There are two seeds per cap cell; open-versus-closed is confounded with model family; and because the cap is a soft request, the open model does not comply with it reliably, so these runs corroborate rather than replicate. The strong evidence is the closed-model pinning, which is visible in the *already-published* run records and requires no new experiment. Settling the question properly requires re-running frontier models at several caps. Until then we report the procedure gains as real and withdraw the claim that they are one-shot.

#### 4.4 T4 — Closed→Open

A closed researcher rewriting an open trainee’s training harness can help substantially (`combined_rsi.json`): Fable-5.1 → Qwen-1.5B reaches improved–frozen **+0.472** (peak 0.472,  $C$  0.147 → 0.501 vs frozen 0.029), and Fable-5.1 → Qwen-7B **+0.313**. But the sign is *pair-dependent*: the same researcher *hurts* other trainees (Fable→OpenCoder-1.5B  $-0.265$ , Fable→DeepSeek-1.3B  $-0.444$ ), and many researcher rows are flat. As with T4’s cousin T2, an open question (ROADMAP A4) is whether the  $+0.472$  is correctness rather than speed.

**Intuition.** *Improvement does transmit through authored tooling — a closed model that cannot change its own weights can still cause a better open model. But it is brittle and trainee-specific, which reads less like “the researcher discovered a better training recipe” and more like “it happened to match this trainee’s correctness headroom.” Transmitted improvement is real but not yet a reliable, general channel.*

#### 4.5 T5 — Self-play: an author-yield failure, not a measured null

The primary signal is  $L-F$  (extra gain from a co-evolving author beyond an escalating curriculum).  $L-F$  is only defined if the live author actually produced accepted, valid, novel tasks. **It did not, at the yield needed to measure anything.**

**Open** (`selfplay.json`, 24 runs): most models  $\leq 15B$  report  $L-F \approx 0$ . Requiring at least five accepted model-authored tasks before a row is interpreted, only 6/24 open runs qualify. **Closed** (`selfplay-closed.json`, procedure channel, 11 runs): 10/11 rows are *exactly*  $L-F = 0.0$  — including runs whose authors proposed 36–37 tasks across 24 rounds. An exact zero across 24

rounds with non-zero proposal counts indicates the co-evolution channel never engaged, not that it engaged and returned nothing. **Open-ended** (`open_rsi.json`, 16 runs): all verdicts are “one-time (open = fixed)”.

**Finding.** The two rows previously quoted as positives do not survive the yield floor: starcoder2-15b  $L-F = +0.232$  rests on **one** accepted authored task over 6 rounds, and deepseek-coder-6.7b-base  $L-F = +0.154$  on **three** over 3 rounds. Their per-round  $L-F$  traces oscillate in sign. We therefore withdraw both from the headline claims and report T5 as *undefined at the current author yield*.

**What the dedicated diagnosis run actually shows.** The self-play diagnosis (`selfplay_diag.json`, Qwen2.5-Coder-3B, 8 rounds) authored **zero** valid tasks in every round. Consequently every per-round diagnostic in that file — authored solve-rate, trivial and unsolvable fractions, authored headroom — is `null`, and all three arms scored  $C=0$ . An earlier draft of this report read that run as showing a *bimodal* authored-difficulty distribution (tasks either trivial or unsolvable, rarely in a learnable band). **That conclusion is not in the data and we retract it:** the bimodality is a plausible hypothesis the run was unable to test, because nothing was authored to measure. The supported statement is narrower: *free-form task proposal collapses*. Without constrained, executable task mutation, the author does not manufacture a frontier, so self-play cannot be tested at all — which makes structured mutation with validity and learnability gates a prerequisite experiment, not an extension.

(Note: an earlier README listed closed Sonnet-5 +0.045 / GPT-5.6-Sol +0.041 as “emerging positives”; these are *not* in the current data and have been removed.)

## 5 Cross-cutting analyses: the mechanism

The rung-by-rung results tell us *whether* improvement compounds. The following analyses tell us *why* it does not, and together they form the mechanistic core of the report.

### 5.1 Coverage vs. scale, and the “correctness wall” as a sampling artifact

Per-task  $p$ -maps on a fixed 29-task bank (`pmap_curve.json`) trace coverage (tasks with  $\geq 1$  verified-correct sample) from 13.8% at 0.5B to 86.2% at 14B, and show that below the wall  $\text{pass}@K$  dwarfs  $\text{pass}@1$ :

| Size | $K$ | coverage | $\text{pass}@1$ | $\text{pass}@K$ | $K$ -ratio      |
|------|-----|----------|-----------------|-----------------|-----------------|
| 0.5B | 64  | 13.8%    | 0.009           | 0.109           | $\sim 12\times$ |
| 1.5B | 64  | 27.6%    | 0.014           | 0.235           | $\sim 17\times$ |
| 3B   | 64  | 82.8%    | 0.112           | 0.777           | $\sim 7\times$  |
| 7B   | 48  | 82.8%    | 0.079           | 0.799           | $\sim 10\times$ |
| 14B  | 24  | 86.2%    | <b>0.655</b>    | 0.862           | $\sim 1\times$  |

Table 3: Coverage and sharpening across scale. The sharpening geometry (`results_auto.tex`): the *sharpenable* band ( $0 < p < 0.2$ ) is 14%/28%/59%/79%/0% at 0.5/1.5/3/7/14B — it peaks mid-scale and collapses at 14B, where all mass is already *reliable* ( $p \geq 0.2$ ).

**Intuition.** *The sub-2B “correctness wall” is not an inability — it is a low per-sample success probability. Give a small model enough draws and correct kernels appear ( $\text{pass}@K \gg \text{pass}@1$ ). This reframes the entire problem: self-training does not need to learn to be correct, it needs to*

reliably reproduce a correctness that is already latent in the sampling tail. That is a sharpening problem, not an exploration problem.

## 5.2 Zero-score forensics: the wall is a formation failure

Classifying *why* each 0-score candidate failed (`forensics_summary.json`) shows the sub-3B wall is a **kernel-formation** wall (no parseable ModelNew), not runnable-but-wrong code: `no_extract` dominates at 0.5B (72%), DeepSeek-1.3B (90%), Yi-1.5B (96%). The failure *locus marches downstream with scale*: incoherence  $\rightarrow$  truncation/syntax (Qwen-7B: 75% syntax)  $\rightarrow$  API-hallucination  $\rightarrow$  wrong-output  $\rightarrow$  correct (Qwen-14B: 65% correct, only 0.2% `no_extract`).

**Intuition.** *Small models do not fail by writing subtly-wrong kernels — they fail by not producing a well-formed kernel at all. That is why coverage is so low and why self-training starves: there is nothing correct to train on. As scale rises the bottleneck migrates from “can it write a kernel” to “is the kernel right” to “is it fast.”*

## 5.3 Search beats training at matched compute

Analysed at the trajectory level over 57 trajectories / 228 round-comparisons (`search_vs_train.json`), lineage—bestof $N = -0.111 [-0.138, -0.083]$  (CI excludes zero; the cluster-robust interval  $[-0.163, -0.120]$  agrees), and verified search beats lineage in **84% of runs** (76% of rounds); every well-powered scale is individually negative (0.5B  $-0.123$ , 1.5B  $-0.114$ , 3B  $-0.123$ ; 7B  $-0.018$  at  $n=5$  is not interpreted). Unlike the compounding null, this positive *strengthens* under the clustering correction, because the contrast is made within a round. A separate matched-budget baseline sweep (`baselines.json`, budget 20) is weaker and mixed — Qwen-1.5B recursion\_gain  $+0.055$  over its best non-recursive baseline, but starcoder2-3b  $-0.073$  — so the strong, consistent statement is the search-vs-lineage one.

**Finding.** At a matched compute ledger, best-of- $N$  verified search beats lineage self-training (lineage—bestof $N = -0.111 [-0.138, -0.083]$  over 57 trajectories, CI excludes zero). The useful solutions are *already reachable by sampling*; search exploits them directly, while training must compress them into shared weights and have the gain survive.

## 5.4 WHY-RSI: the weight-level mechanism and the inverted-U in scale

Across 133 weight-RSI runs (`mech_analysis.json`) the correctness wall is the primary gate: crossing rises from 54% below 2B to 92% at  $\geq 2B$ . Among the 97 wall-crossers, RSI is an **inverted-U in scale**:

| Band      | $n$ | wall-cross | LoRA drift   | retention | frac RSI    |
|-----------|-----|------------|--------------|-----------|-------------|
| <2B       | 67  | 0.54       | 0.019        | 0.013     | 0.03        |
| 2–8B      | 45  | 0.98       | 0.246        | 0.180     | <b>0.49</b> |
| $\geq 9B$ | 21  | 0.81       | <b>0.726</b> | 0.339     | 0.19        |

Table 4: The inverted-U (`mech_analysis.json`,  $n=133$ ). RSI peaks at 2–8B; the largest models drift *most* in weight space yet compound *least* — motion without progress near the roofline. 28/133 runs met the compounding criterion (stable-code-3b recurs). When drift occurs it localizes to *late* layers (task adaptation) and the held-family transfer gap is  $\approx 0$  (no transferable meta-skill).

**Finding.** A subtle and important nuance: on this bank, compounders have *lower* generation diversity than flat runs (0.41 vs 0.52), not higher. The discriminator between compounding and flat among wall-crossers is **sustained LoRA drift + retention** ( $\sim 5\times$  higher in compounders), *not* diversity preservation. So “diversity collapse causes the null” is *not* supported by our weight-RSI data — compounding here looks like *productive convergence onto a good basin*. Diversity contraction remains a plausible *cross-round option-value* lock (see the Discussion) but we flag it as an association whose clean test is a diversity-preserving intervention, and we explicitly do *not* claim it as the mechanism.

**Intuition.** *Scale does not monotonically help RSI — it is an inverted-U. Small models cannot form correct kernels often enough to feed the training set (formation-starved). Mid-scale models sit on the widest learnable frontier and compound best. Large models cover almost everything already, so their weights churn (high LoRA drift) without moving the held-out score — “motion without progress” near the roofline. The binding constraint slides from formation (small) to headroom (large).*

## 5.5 Probe-as-intervention: correctness is represented better than it is generated

A linear probe reads “is this kernel correct” out of mid-to-late layers at high AUC (Qwen-3B best\_auc **0.985** at depth 0.86, while its actual correctness rate is 0.105) — the model *represents* correctness far better than it *generates* it (`interp.json`).

**Finding.** Analysed at the right unit, probe-guided selection *does* beat uniform-random selection at matched *generation* budget. Clustering the 19 usable runs on their 12 distinct base checkpoints (rows sharing a checkpoint are repeated draws, not distinct models), the mean lift is **+0.124** [**+0.029**, **+0.218**] — the 95% CI excludes zero — recovering 36% [11%, 60%] of the oracle–random headroom. An earlier draft read the same rows as “no advantage” because it looked at the spread of individual runs (−0.27 to +0.45) rather than the clustered mean. See Appendix A.

**Three reasons this is a modest secondary result, not a headline.** (i) *Uniform random is the weakest possible comparator*: the informative ones — length-normalised log-likelihood reranking, a compile/run validity filter, execution agreement — were not run, so we cannot say the probe is the best use of the same compute. (ii) *The denominators are tiny*: each run’s lift is computed over 5–12 test tasks, so a “+0.45” is about five problems, and the clustered interval is correspondingly wide. (iii) *Matched-generation selection is not matched-verification harvesting*: a gain in top-1 correct rate from reranking an existing pool does not imply the probe makes a verification-constrained search cheaper.

**What we cannot verify from the released artifacts.** Two figures quoted in earlier drafts — a within-task ranking AUC of  $\approx 0.67$  and a matched-verification harvesting yield of +0.01 — come from a probe-baselines run whose per-candidate outputs are *not* in this repository. Per-candidate probe scores were not retained anywhere, so the within-task AUC cannot be recomputed from what we release. We therefore do not rely on either number, and we state the within-task estimand as **unmeasured** rather than as measured-and-small. This is the single most important gap in the probe story: only a within-task ranking metric can explain a selection gain mechanistically, and the pooled AUC cannot substitute for it — as the runs with pooled AUC=1.000 and lifts anywhere from +0.000 to +0.250 demonstrate.

**Intuition.** *A linear probe can read “is this kernel correct” out of mid-to-late layers at high AUC, yet the model emits a correct kernel far less often than it can internally recognize one. The model*

| Defect                                                        | Presented as                             | Caught by                                                       |
|---------------------------------------------------------------|------------------------------------------|-----------------------------------------------------------------|
| Grader mis-imported under a wrong root                        | “self-training does not com-pound”       | identity-kernel assertion                                       |
| Grading device index absent on a 1–2 GPU allocation           | “the model writes bad kernels”           | same assertion, run <i>on the allocation shape the jobs use</i> |
| Raw generations graded without extracting the submission      | “the 14B teacher solves 0/29”            | non-zero coverage precondition                                  |
| Score saturated at the correctness floor                      | “lineage $\approx$ reset”                | score-histogram check                                           |
| Compiled-baseline cache hitting a filesystem inode quota      | “4 of 6 rounds produced nothing”         | output-path/quota correlation                                   |
| Our own explanation of the first five                         | “H100 closed the headroom”               | tracing prompt $\rightarrow$ number                             |
| Host CC leaking into the container, so Triton could not build | “models cannot write kernels when asked” | known-good <i>Triton</i> kernel                                 |

Table 5: Seven defects, each of which independently manufactured a null. The middle column is what the artifact would have claimed had the defect not been found. Every one of these is a statement we came close to publishing.

knows *more than it generates*. *This is the same discovery $\rightarrow$ assimilation gap seen from the inside: the limiting step is turning latent competence into a reliably-produced output.*

## 6 Instrument validity: how a self-improvement benchmark manufactures a null

A benchmark that reports a null owes its reader a second result: evidence that the measurement *could have* reported a positive. Self-improvement benchmarks are unusually exposed here, because the headline quantity is a difference between two arms of the same pipeline. Any defect that suppresses both arms equally leaves the contrast at zero while every surface diagnostic — jobs complete, kernels compile, scores populate — looks healthy. A broken instrument does not report that it is broken; it reports  $\Delta \approx 0$ .

Over this project, **seven independent pipeline defects each produced a plausible, publishable-looking null**, summarised in Table 5. None announced itself, and three survived a code review before being caught by an assertion. Six were found only after they had been written up as findings; one was caught before it reached prose, and we report what made the difference.

**A clean zero is a bug hypothesis before it is a finding.** The operational signature that caught most of these is distributional, not statistical. A *real* null is noisy: it has per-task spread, partial successes, seed-to-seed variance. A null in which every task fails identically, or every score takes the same value, is instrumentation. We therefore treat suspicious tidiness as evidence *against* the result, inverting the usual intuition that clean data is good data.

**Correctness of the instrument is not sufficient; it must also have range.** This is the failure mode we consider most transferable, because it defeats the obvious check. Our headroom-normalised score is  $0.5 + 0.5 \cdot \text{clamp}\left(\frac{sp-1}{\text{ceiling}-1}\right)$ , which is well-formed, was verified end-to-end on known-good inputs, and was working exactly as specified. It was nonetheless useless as a contrast:

on H100, **76% of all scores across every run and round were exactly** 0.50 — precisely the value of a correct submission at parity with the baseline. With both arms pinned to the same absorbing value, lineage — reset is zero *by construction*, and the experiment could not have detected compounding in either direction.

**The sixth defect was our own explanation of the first five.** Our initial diagnosis of that spike was that the `torch.compile` baseline is relatively stronger on H100, closing headroom that existed on A100. It was plausible, it fit the data, and it was wrong. The default scorer does not use the compiled baseline at all: it evaluates speedup against *eager* PyTorch with the legacy fixed 1.5× anchor. No claim about `torch.compile` was licensed by those runs, because `torch.compile` never entered the measurement.

The actual cause was upstream of the metric, in the prompt. Our generation instruction contains the clause “*a plain-torch kernel that is correct beats a fancy one that errors*” — written to suppress parse and compile failures, which it did. Of **86 verified kernels** harvested from the 14B model, **zero** contained Triton, CUDA, or `load_inline`. All 86 were pure-PyTorch rewrites of the reference module. An independent re-harvest under the same prompt replicated this exactly: **0 of 118 verified kernels**, for **0 of 204** across both. A semantically equivalent rewrite runs at eager speed by construction: median speedup 1.00×, with 93% below 1.05×. And `_score(correct, 1.00) = 0.50` exactly.

So the chain is deterministic, and it runs entirely through our own harness: a prompt that discourages optimisation → models that decline to optimise → speedups pinned at 1.00× → a score pinned at 0.50 → a contrast pinned at zero. The metric was never broken. It reported, accurately, that the models were not attempting the task the benchmark exists to measure — and we misread a correct measurement as a broken instrument for long enough to write the wrong explanation down.

We record this as the most instructive failure in the project, because it inverts the lesson of the previous five. There, a clean zero was a bug. Here, a clean zero was a *finding*, and the bug was in the interpretation. The usable rule is therefore not “suspect the instrument” but the stricter **trace the causal chain from prompt to number before naming a cause**, and prefer the explanation that is testable by intervention: ours was settled by an A/B on the prompt text, holding tasks, grader, model and seed fixed.<sup>1</sup>

Critically, none of this is repairable by task curation. Filtering the bank against an H100 anchor with a 1.3× minimum roofline headroom kept 23 of 29 tasks and still returned a frozen-base mean best score of 0.511, with every surviving task at  $\approx 0.50$ . A filter selects among existing tasks; it cannot make a model attempt an optimisation it was steered away from.

**The estimator, not the normalisation, is the thing to fix.** A saturated metric invites the wrong repair. The instinct is to re-normalise; the actual problem is that best-of- $k$  over a speed score is the wrong *estimator* for a population sitting at the correctness floor. Best-of- $k$  answers “can it ever,” and returns an identical 0.500 whether a model solves a task once in eight attempts or eight times out of eight — two plainly different models. Where the live axis is reliability rather than speed, pass-rate over  $k$  separates those cases by a factor of eight in resolution.

We report such runs under a separate metric (pre-registration, Amendment 1), never pooled with the headroom leaderboards, and with two properties stated up front: the denominator is  $k$ , so

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<sup>1</sup>That A/B is running at submission time and its outcome is reported in the camera-ready. Until it lands, “models cannot beat the baseline on this hardware” and “models were told not to try” are observationally identical, and only the second is actionable — which is exactly why the distinction has to be drawn before either is asserted.

a model emitting nothing parseable scores zero rather than being excluded; and it does not reward speed, so it tests a *weaker and different* claim than the headroom score. It was also adopted *after* observing saturation — a forking path that we record as such rather than present as the original design.

**Check the denominator of the mechanism, not just the metric.** The same failure shape recurs wherever a rate is computed over events that never occurred. Two frontier models in our procedure-RSI track recorded  $n_{\text{strategies}} = 0$  in every round: their output was never parsed into a strategy list, so the self-modification channel never engaged. The resulting  $-0.234$  “self-degradation” was widely quoted from an earlier draft of this work and is a parse failure. Likewise, our self-play primary metric requires the live author to produce accepted valid tasks; demanding at least five qualifies only 6 of 24 open and 5 of 11 closed runs. **A metric computed over zero events is undefined, not zero**, and the distinction is invisible in the aggregate.

**The seventh defect: the benchmark had never measured its own subject.** Having traced the plateau to the prompt, we tested the obvious follow-up — remove the steer and ask explicitly for a Triton kernel. The model solved **0 of 29** tasks, against 25 of 29 under the published prompt. That is a clean, pre-registered, publishable result: the models try when asked and cannot produce a working kernel.

It is also false. Before reporting it we applied the known-good-input gate to the *new* input: we hand-wrote a Triton kernel that is correct by construction and required the real grader to verify it, alongside a plain-torch control that must pass and a wrong kernel that must fail. The controls behaved; the correct Triton kernel returned `FileNotFoundError`. The cause was that Triton compiles a small C shim at first use and takes its compiler from `$CC`, which the cluster’s module system sets to a *host* toolchain path that does not exist inside our container. Plain PyTorch never compiles anything at runtime, so **only Triton broke** — and Triton is the subject of the benchmark. One environment variable fixed it, after which the same kernel verifies.

The consequence is not a corrected number but a corrected scope: for the entire history of this project, **every submission that was not plain PyTorch failed for an environment reason**. The benchmark has been measuring correctness-preserving rewriting, not kernel optimisation.

This also revises our own account of the previous defect. We attributed the saturated score to the prompt’s instruction to prefer plain torch — but that instruction now reads as a *rational adaptation to a broken grader*. Tuned against a harness in which Triton always failed, plain torch genuinely was the only thing that worked. The prompt and the grader were coupled, and the grader came first. Generalising: **a configuration that looks inexplicably conservative may be compensating for a defect**, and because the defect never appears in any result, the workaround looks like a design choice indefinitely.

Two things made this one catchable before publication rather than after. First, failure reasons now propagate: the grader had always computed a reason string and discarded it one line later, so every failure in project history was a bare `False` — and “model writes Triton, Triton grades `False`” reads naturally as a statement about the model. Second, and more transferable: **when an experiment begins producing a new kind of artifact, the known-good gate must be re-instantiated for that kind**. “The grader works” was true and useless; “the grader works on Triton” was the question, and nothing was asking it.

**What the corrected instrument then showed.** With Triton verifying, the prompt A/B could finally be run as intended. Classifying every generation (rather than counting only successes) across

four tasks at  $k=4$ ,  $n=16$  per arm:

| Prompt    | Outcome                          | $n/16$    | 95% CI      |
|-----------|----------------------------------|-----------|-------------|
| published | plain-torch rewrite              | 16        | [81%, 100%] |
| published | attempted a custom kernel        | <b>0</b>  | [0%, 19%]   |
| kernel    | attempted a custom kernel        | <b>14</b> | [64%, 97%]  |
| kernel    | of which <i>verified correct</i> | <b>1</b>  | —           |
| kernel    | of which failed verification     | 13        | —           |

This separates two quantities the project had conflated throughout. **The prompt controls whether the model attempts a kernel at all** — 0 attempts versus 14 is an instruction being followed, not a capability difference. **Capability controls whether the attempt succeeds** — 1 of 14 verified. The model can write a working Triton kernel and usually does not. With  $n=16$  the interval is wide and the point estimate is not the claim; the claim is that the rate is *neither zero nor high*.

That distinction is the most consequential measurement in this report, because it explains every saturation we chased. The 0.50 floor, the ceiling at 1.000, and a coverage-injection control that exhausts its task set by round 2 all share one cause: **these models have already saturated the task the benchmark was actually measuring**, which is correctness-preserving rewriting rather than kernel optimisation. Any metric built on a saturated task piles up against one wall or the other, and we tried three — a headroom score, a pass-rate score, and a 456-task generated bank — before recognising that.

**The substrate problem was therefore never a metric problem.** The remedy is not a better scoring rule but a harder task, and the harder task only became measurable once the grader could verify one. We regard this as the central practical lesson of the work: when a population sits against a wall under every metric you try, the task is too easy, and no amount of re-normalising will manufacture the variance that is missing.

**Reachability: the dynamic-range precondition, made quantitative.** “Check the metric has range” is advice until it is a number, so we propose one. **Reachability** is the fraction of bank tasks on which the *frozen base* already produces a submission clearing the scoring anchor. It is cheap — it falls out of the same calibration pass used to admit tasks — and it is computed before any experiment runs.

Measured on the two banks we calibrated against an identical H100 anchor:

| Bank                     | Tasks | Admitted | Reachability     |
|--------------------------|-------|----------|------------------|
| Hand-curated (29-task)   | 29    | 23       | 1 / 29 (3.4%)    |
| Generated DSL (456-task) | 456   | 180      | 57 / 456 (12.5%) |

Table 6: Reachability = fraction of tasks where the frozen base clears the scoring anchor (1.5× over eager). Both banks are bimodal — parity, or well past the anchor — but they differ four-fold in whether the speed dimension is reachable at all.

A reachability of 3.4% means that on 28 of 29 tasks, *every* success the base produced landed at parity, so the score had one attainable non-zero value and the contrast between two arms was pinned before a single round ran. We would now decline to run a speed-scored compounding study on such a bank, and we report the statistic for any bank we publish.

Note what this does *not* say. The measurement above is against eager PyTorch at a fixed  $1.5\times$  anchor, which is what the default scorer uses; the roofline ceiling does not enter it. It therefore licenses no claim about `torch.compile`, and a bank of *generated* references may be easy to beat for reasons that do not transfer — a naive reference is not a strong baseline. Reachability is a necessary condition for a usable substrate, not a sufficient one.

**What we propose as standard practice.** We ship these as executable gates rather than advice, and we recommend them for any self-improvement benchmark reporting a null:

1. **Known-good input, known-good output**, asserted before every batch and run *on the same allocation shape as the experiment* — one of our defects lived entirely in the difference between a 1-GPU and a 3-GPU allocation.
2. **A dynamic-range precondition**, reported as *reachability* (Table 6): the fraction of tasks on which the frozen base already clears the scoring anchor. Publish it beside every bank, and never interpret a contrast computed on a bank where it is near zero.
3. **A positive control on the causal path**: an intervention known to move the primary metric (for us, injecting verified solutions on tasks the student failed). A null is interpretable only once the harness has been shown to register a positive.
4. **Mechanism denominators reported alongside every rate**, so undefined is distinguishable from zero.
5. **Automated cross-artifact consistency checking**: every number in the manuscript re-derived from the data files at build time. Ours catches stale figures and claim strings that contradict the data they cite — both of which occurred.

We emphasise that this is not a catalogue of unusual carelessness. Each defect was a single line, each was locally reasonable, and each was caught only because an assertion was looking for it. The generalisable claim is structural: **in a benchmark whose headline is a difference between two arms of one pipeline, the cheapest result to produce accidentally is a null**, and the field currently has no convention requiring authors to rule that out.

## 7 Synthesis: why self-improvement does not compound here

### 7.1 Why self-improvement does not compound

**The mechanism: a discovery→assimilation bottleneck (sharpening, not exploration).** Our two strongest results jointly localise the failure. Best-of- $N$  search beating lineage training at matched compute (lineage–bestof $N = -0.111 [-0.138, -0.083]$  over 57 independent trajectories), together with the large  $\text{pass}@K \gg \text{pass}@1$  gap, shows that *useful solutions are already present in the sampling tail* but are not reliably produced in one sample. The limiting pipeline is therefore

rare verified sample  $\rightarrow$  selected SFT example  $\rightarrow$  weight update  $\rightarrow$  reliably reusable gain,

and it is *lossy*: search can exploit a rare discovery directly, whereas training must compress it into a shared parameterisation, preserve other skills, and have the gain survive later rounds. The lineage-vs-reset null (inheriting updates does not beat restarting) is direct evidence *against a useful cumulative learning state*, not merely evidence that individual updates are small. This is the classic self-training signature: selected-sample training concentrates probability on *already-reachable* successes faster than it expands what is reachable (cf. STaR, ReST, ReST<sup>EM</sup>, expert iteration; and the RLVR capability-boundary debate, Yue et al. 2025).

**A formal fixed point — with its limits stated precisely.** Under an *idealised* support-preserving rejection-conditioning operator  $R$  (unlimited accepted samples, exact per-task fitting, no cross-task transfer), exact solution-support coverage  $C_0(\pi) = \Pr_x[p_\pi(x) > 0]$  is invariant and  $R$  is idempotent after one step:  $C_0(R\pi) = C_0(\pi)$  and  $R^2\pi = R\pi$ . On tasks with positive success probability  $R$  places all mass on accepted outputs; on  $p=0$  tasks no accepted distribution exists. So an idealised verified-SFT loop is a *one-step fixed point in exact coverage*. **We are careful about what this does *not* say.** (i) Neural SFT is not this operator — shared parameters can generalise and raise the probability of unsampled outputs, so real SFT need not preserve support. (ii) *Finite-budget* coverage is *not* invariant:  $C_K(\pi) = \mathbb{E}_x[1 - (1 - p_\pi(x))^K]$  rises sharply under sharpening even when  $C_0$  is fixed — “pass@ $K$  coverage is a fixed point” is false. (iii) Zero observed successes only bound  $p \lesssim 3/K$ ; our “unreachable mass” is more precisely *undiscovered mass at the evaluated budget*. (iv) A training-operator fixed point is *not* a parameter-space local optimum. We therefore report an **operational self-training plateau / protocol-relative fixed point**, never an unqualified “local optimum.”

**Why verified search escapes where training does not.** Search does not escape the policy’s support; it escapes the *one-sample* bottleneck and avoids the assimilation bottleneck. For success probability  $p$ ,  $P(\text{hit in } N) = 1 - (1 - p)^N$  makes extra draws valuable when  $p$  is small, and the verifier keeps the winner. Search (a) uses a discovery directly rather than compressing it into weights, (b) allocates candidates per-task rather than via one shared update, and (c) does not permanently contract the proposal policy. The precise statement: *verified search accesses useful existing tail capability more efficiently than the tested training loop consolidates it*. (This does not settle lifetime economics — a training update could amortise over enough future queries; that needs a separate deployment-horizon study.)

**Diversity contraction as a reinforcing lock (associational).** Selected-sample SFT can drive a loop: current common winners  $\rightarrow$  accepted set  $\rightarrow$  higher probability on those winners  $\rightarrow$  fewer alternatives in the next accepted set. The loss is *option value* — rare decompositions and kernel families that would enable *later* discoveries. If undiscovered useful mass  $U_t$  shrinks, later rounds find fewer novel successes  $(1 - (1 - U_t)^N \leq NU_t)$  even as pass@1 rises on familiar ones. Higher retention among compounders and diversity collapse at large scale are *consistent* with this lock, but we flag them as association, not proof: low diversity can also reflect successful specialisation, and the cleanest test is a matched-reward, matched-example, diversity-*preserving* intervention.

**The inverted-U, mechanistically.** Partition tasks into *undiscovered* ( $p \approx 0$  at budget), *learnable frontier* ( $0 < p < 0.2$ ), and *near-ceiling* ( $p$  high). Most self-training value comes from the middle band. Small models: formation failures ( $P(A | x) = P(F | x)P(A | F, x)$  with tiny  $P(F)$ ) starve the accepted set; mid-scale: the frontier is widest, so discoveries become signal (RSI peaks); large scale: high coverage but little remaining headroom, so drift redistributes mass within already-covered behaviour (“churn without gain” — supported descriptively; “only superficial late-layer edits” is *not* yet established, as parameter distance is not calibrated functional change).

**What would break the plateau, and what our data predict.** Two distinct breakouts are needed: *accumulation* (retain/express more already-discoverable successes — could yield lineage>reset) and *frontier expansion* (make new strategies operationally reachable — required once the frontier is exhausted). A productive loop needs new discoveries + assimilation + retention + headroom simultaneously. The single most informative next experiment is a **coverage-injection**  $\times$  **diversity-**

**preserving** factorial: inject stronger-model / repair / search-generated solutions on *undiscovered* tasks and test whether inherited gains then emerge — separating “lack of discoveries” from “loss of discoveries after learning.” Exploration bonuses, novelty-selected accepted sets, prerequisite curricula, off-policy replay, and search-amplified expert iteration are ranked candidates; each helps only the specific deficit it targets (e.g. entropy cannot fix formation; off-policy replay of redundant winners adds no frontier information).

**Scope — claims we do *not* make.** We do not claim the models’ reachable set is mathematically invariant, that the weights sit at a genuine local optimum, or that RL/RLVR cannot expand capability. We claim, with *moderate* evidence for the null and a significant positive respectively: under verified rejection-sampling SFT at the tested scales/budget, carrying the lineage forward does not beat a matched reset (trajectory-level TOST equivalence at  $\delta=0.05$ ,  $\text{BF}_{01}=5.6$  over 57 independent trajectories); verified search is a better use of the same compute (CI excludes zero); and the binding internal signals are coverage/formation (small), a mid-scale learnable frontier, and drift-without-gain plus diversity contraction (large). The synthesis: *expert iteration without sufficient expert amplification, plus incomplete retention of the apprentice’s useful tail.*

## 8 The accumulated intuition (what three weeks taught us)

This section is the part that does not fit in a conference paper: the qualitative model we now carry in our heads about RSI in this domain.

1. **Self-training is a sharpening operator, not an exploration operator.** It concentrates probability on already-reachable successes faster than it expands what is reachable. Everything else follows from this.
2. **Coverage is the currency, and it is not self-generated.** A round of self-training can only reuse solutions the model already finds; it cannot manufacture coverage on tasks where the per-sample success probability is effectively zero at the evaluated budget. Compounding is coverage-limited, and coverage is exogenous to the loop.
3. **The bottleneck is discovery→assimilation, not knowledge.** The model represents correctness it cannot reliably generate (probe result), and search — which uses discoveries directly — beats training, which must assimilate them into weights and retain them. The lossy step is assimilation-and-retention.
4. **“Operational plateau,” not “local optimum.”** We are careful: the null is a protocol-relative fixed point of the tested verified-SFT loop, not a claim that the weights sit at a parameter-space local optimum. Finite-budget coverage is *not* invariant; a better loop could still move.
5. **Diversity contraction is a plausible reinforcing lock.** Selected-sample SFT can raise probability on current winners at the cost of the rare decompositions that would enable *later* discoveries. We flag this as association, not proof — the clean test is a diversity-preserving intervention.
6. **The open contribution is a better *harness*, not a better model.** Because RSI is a protocol we run, the way to break the plateau is to change the protocol: inject coverage on undiscovered tasks, preserve diversity in the accepted set, amplify the “expert” with search

before assimilation. This is exactly what the public benchmark’s harness-submission track exists to surface.

## 9 What would break the plateau: predictions

Two distinct breakouts are needed and they target different deficits: *accumulation* (retain and express more already-discoverable successes — could make lineage finally beat reset) and *frontier expansion* (make genuinely new strategies operationally reachable — required once the current frontier is exhausted). A productive loop needs new discoveries + assimilation + retention + headroom *simultaneously*. Our single most informative next experiment was, at the time of writing, a **coverage-injection**  $\times$  **diversity-preserving** factorial. The instrument work in Section 6 has since superseded that. The coverage-injection control was run, and its injected-task count falls to zero by round 2 because the student solves the entire held set; injection cannot separate “lack of discoveries” from “loss of discoveries” on a task the model has already saturated.

The informative experiment is now the one a repaired grader makes possible for the first time: run the compounding protocol on **kernel authoring** rather than correctness-preserving rewriting. Under the published prompt, models attempt a custom kernel in **0 of 16** samples; when asked explicitly, **14 of 16**, of which **1** verifies. A success rate in that band is far from both the floor and the ceiling that defeated every metric we tried, which is precisely the regime a compounding study requires. Whether self-training compounds on a task the model has *not* saturated is, as far as we can tell, still open — and it is a question this benchmark could not previously have asked.

## 10 Limitations and scope

We claim, with strong evidence: under verified rejection-sampling SFT at the tested scales and budgets, carrying the lineage forward does not beat a matched reset; verified search is a better use of the same compute; and the binding internal signals are coverage/formation (small scale), a mid-scale learnable frontier, and drift-without-gain plus diversity contraction (large scale). We do *not* claim the reachable set is mathematically invariant, that the weights sit at a genuine local optimum, or that RL/RLVR (as opposed to rejection-sampling SFT) cannot expand capability. All primary intervals are now computed at the *trajectory* level rather than the round level, since rounds within a run share a base checkpoint, a seed, a held-out split and an accumulated adapter; this is the honest unit of replication and it costs us evidential strength, moving the equivalence Bayes factor from  $BF_{01} = 14.7$  (pooling 228 rounds) to 5.6 (57 trajectories) — strong to *moderate*. The verdicts survive the correction, and the search-beats-training positive strengthens under it, because that contrast is made within a round. Per-trajectory (2–3 seed) uncertainty remains the binding constraint. Finally, all results here are A100; our H100 runs are reported separately and are non-comparable on two axes at once (different roofline constants, and a `torch.compile` baseline strong enough to close the measurement range entirely).

## 11 Conclusion

Three weeks of clean-verifier, compute-matched experiments across five rungs of recursion and scales from 0.5B to 14B converge on one picture: in GPU-kernel optimization, verified self-training *sharpens what a model already covers but does not explore*, so improvement does not compound — not because the models lack capability, but because the loop’s feedback channel assimilates rare discoveries lossily and cannot manufacture the coverage it would need to keep going. The result is

a strong, honest null flanked by two positive findings (search beats training; the correctness wall is a sampling artifact) that together localize the failure and point at the intervention most likely to break it. KernelAscent ships that intervention as an open, automated submission track, so the community can try to turn the null into a curve.

## A Appendix: probe-as-intervention, analysed as a secondary result

A linear probe reads “is this kernel correct” out of mid-to-late layers at high pooled AUC. This appendix asks whether that decodable signal converts into a *selection* advantage, and reports it at the right unit with the right denominator.

**Three corrections to the earlier analysis.** (1) **Unit.** The 19 usable runs are repeated probe-training/test draws over **12 distinct base checkpoints**; draws sharing a checkpoint are not independent, so we cluster on checkpoint. (2) **Denominator.** Each run’s lift is computed over  $n_{\text{test}}$  tasks, and  $n_{\text{test}} \in [5, 12]$  (median 12). A “+0.45 lift” is roughly five problems, so  $n_{\text{test}}$  is printed in every row. (3) **Estimand.** A high *pooled* correctness AUC can arise because the probe separates easy tasks from hard ones while failing to rank candidates *within* a task — and only within-task ranking can produce a selection gain. Per-candidate probe scores were not retained, so the within-task AUC **cannot be recomputed** from the stored artifacts. We flag this as the binding limitation rather than letting the pooled number stand in for it.

**Result.** Clustered on base checkpoint, the mean lift over uniform-random selection is +0.124 [+0.029, +0.218] over  $n=12$  checkpoints — the 95% CI **excludes zero**: against a uniform-random selector at matched generation budget, probe-guided top-1 selection does lift the correct rate. The result is also *not* TOST-equivalent to zero at the pre-declared  $\delta=0.05$ , so neither a benefit nor its absence would have been concluded by default — the interval is simply wide. The naive row-level pooling would report +0.118 [+0.028, +0.209] over 19 rows. The probe recovers 36% of the oracle–random headroom on average (95% CI [11%, 60%]).

**What this does and does not establish.** **Uniform random is the weakest possible comparator.** Beating it shows the probe carries *some* usable signal; it does not show the probe is the best use of the same compute, because the informative comparators were not run: mean/length-normalised log-likelihood reranking, a compile-or-run validity filter, and execution agreement. Separately, and importantly, this matched-*generation* selection gain is a different quantity from matched-*verification* harvesting yield, where we measure no advantage. The coherent joint statement is: *extra draws plus a cheap selector find more correct kernels, but the tested probe does not make verification-constrained search economically better.*

**The diagnostic.** Several runs achieve pooled AUC=1.000 with lifts from +0.000 to +0.250, on training sets as small as 2–5 correct examples. Perfect pooled separation that does not predict selection gain is direct evidence the pooled metric partly measures task difficulty rather than within-task candidate quality — a trap worth flagging for anyone reading correctness probes off a small bank.

**Defensible claim.** *Decodable correctness information yields a measurable but modest gain over random selection, and does not automatically translate into useful search efficiency under a verification budget.* Settling it needs a rerun on a family-split bank of  $\geq 300$  independent tasks that logs per-candidate scores, with leave-task-out probe fitting, matched-candidate comparators, task-clustered uncertainty, and a full generation/scoring/verification cost ledger.

Table 7: Probe-as-intervention, one row per run (not per model).  $n_t$  = test tasks the row is computed on; head% = fraction of the oracle–random headroom recovered. Rows sharing a base checkpoint are repeated draws and are clustered in the analysis.

| run           | $n_t$ | $K$ | AUC   | random | probe | lift   | head% |
|---------------|-------|-----|-------|--------|-------|--------|-------|
| qwen05k32     | 12    | 32  | 0.831 | 0.218  | 0.667 | +0.449 | 64%   |
| qwen05k32b    | 12    | 32  | 0.878 | 0.247  | 0.667 | +0.420 | 63%   |
| qwen05_nt24   | 5     | 32  | 0.789 | 0.086  | 0.000 | -0.086 | -75%  |
| Qwen1.5_K32   | 12    | 32  | 0.973 | 0.040  | 0.333 | +0.293 | 64%   |
| qwen15k32     | 11    | 32  | 0.961 | 0.337  | 0.545 | +0.208 | 43%   |
| qwen15_nt24   | 5     | 32  | 0.806 | 0.050  | 0.200 | +0.150 | 100%  |
| qwen14k16     | 12    | 16  | 0.872 | 0.698  | 0.750 | +0.052 | 17%   |
| Qwen3         | 12    | 16  | 0.958 | 0.266  | 0.667 | +0.401 | 62%   |
| qwen3_nt24    | 5     | 32  | 0.713 | 0.044  | 0.200 | +0.156 | 100%  |
| qwen3k32b     | 12    | 32  | 0.817 | 0.401  | 0.417 | +0.016 | 3%    |
| qwen3_wt      | 12    | 32  | 0.804 | 0.299  | 0.250 | -0.049 | -11%  |
| qwen7_nt16    | 12    | 32  | 0.824 | 0.531  | 0.667 | +0.136 | 62%   |
| qwen7k32      | 12    | 32  | 0.877 | 0.652  | 0.750 | +0.098 | 28%   |
| qwen7_nt20    | 9     | 32  | 0.819 | 0.379  | 0.111 | -0.268 | -151% |
| SmolLM1.7_K32 | 12    | 32  | 1.000 | 0.083  | 0.333 | +0.250 | 100%  |
| Yi1.5_K32     | 10    | 32  | 1.000 | 0.100  | 0.100 | +0.000 | –     |
| dsc13k32      | 9     | 32  | 1.000 | 0.178  | 0.222 | +0.044 | 28%   |
| DeepSeek6.7B  | 11    | 16  | 0.986 | 0.121  | 0.182 | +0.061 | 40%   |
| dsc67k16      | 12    | 16  | 0.826 | 0.502  | 0.417 | -0.085 | -17%  |